DengueX: AI-Powered Monitoring Platform

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Fall 2025

A Dissertation Submitted To

Faculty of Computing,

Riphah International University, Islamabad

As a Partial Fulfilment of the Requirement for the Award of the

Degree of

Bachelors of Science in Computer Science

Faculty of Computing
Riphah International University, Islamabad

Date: 29/04/2026

Final Approval

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Declaration

We hereby declare that this document **DengueX: AI Powered Monitoring Platform** neither as a whole nor as a part has been copied out from any source. It is further declared that we have done this project with the accompanying report entirely on the basis of our personal efforts, under the proficient guidance of our teachers, especially our supervisor **Mr. Ihtisham Ullah.** If any part of the system is proved to be copied out from any source or found to be reproduction of any project from anywhere else, we shall stand by the consequences.

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Dedication

We dedicate this project to our beloved **Parents**, whose unwavering support, sacrifices, and encouragement have been the foundation of our journey. Their belief in us gave us the strength to overcome challenges and strive for excellence. We also extend our sincere gratitude to our respected supervisor, **Mr. Ihtisham Ullah**, whose continuous guidance, patience, and invaluable insights have played a crucial role in the successful completion of this project. His mentorship has been a constant source of motivation and inspiration. Without the support and encouragement of these individuals, this accomplishment would not have been possible. We are deeply grateful and thankful to them for being an essential part of our academic journey.

Acknowledgement

First of all, we are obliged to Allah Almighty the Merciful, the Beneficent and the source of all Knowledge, for granting us the courage and knowledge to complete this Project.

We would like to extend our sincere appreciation to our respected supervisor, Mr. Ihtisham Ullah, for his continuous guidance, valuable suggestions, and unwavering support throughout the course of this work. His insightful feedback and encouragement played a vital role in shaping this project.

We are also thankful to the faculty members for their support and cooperation, which contributed significantly to our learning experience.

Lastly, we express our heartfelt gratitude to our parents for their endless love, prayers, and encouragement, which have been a constant source of motivation throughout this journey.

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Table of Contents

Table of Contents i

List of Tables iv

List of Figures v

Abstract 1

Chapter 1: Introduction 2

1.1 Introduction 3

1.2 Goal and Objectives 3

1.3 Scope of the Project 4

1.4 Conclusion 4

Chapter 2: Literature Review 5

2.1 Introduction 6

2.2 Background and Problem Elaboration 6

2.3 Detailed Literature Review 6

2.3.1 Definitions 6

2.3.2 Related Research Work 1 7

2.3.3 Related Research Work 2 7

2.3.4 Related Research Work 3 7

2.4 Literature Review Summary Table 8

2.5 Research Gap 8

2.6 Problem Statement 9

2.7 Conclusion 9

Chapter 3: Requirements and Design 10

3.1 Introduction 11

3.2 Requirements 11

3.2.1 Functional Requirements 11

3.2.2 Hardware and Software Requirements 12
3.2.3 Proposed Methodology 13
3.3 System Architecture 13
3.3.1 System Architecture Diagram 14
3.4 Use Cases 15
3.4.1 User Authentication 16
3.4.2 Mosquito Image Detection 17
3.4.3 Breeding Site Reporting 18
3.4.4 Analytics Dashboard 18
3.4.5 AI Chatbot 19
3.5 Sequence Diagram 19
3.6 Component Diagram 20
3.7 Entity Relationship Diagram 21
3.8 Graphical User Interfaces 22
Chapter 4: Implementation and Test Cases 26
4.1 Introduction 27
4.2 Implementation 27
4.2.1 Implementation of Mosquito Detection Module 27
4.2.2 Implementation of User Authentication Module 28
4.2.3 Implementation of Breeding Site Reporting Module 28
4.2.4 Implementation of Analytics Dashboard 28
4.2.5 Implementation of AI Chatbot Module 29
4.2.6 Implementation of News and Updates Module 29
4.3 Test Case Design and Description 29
4.3.1 User Authentication Module Test Case 30
4.3.2 Mosquito Detection Module Test Case 30

4.3.3 Breeding Site Reporting Module Test Case 31
4.3.4 Analytical Dashboard Test Case 32
4.3.5 AI Chatbot Module Test Case 33
4.4 Test Metrics 33
4.4.1 Mosquito Detection Metrics 33
4.4.2 AI Chatbot Metrics 34
4.4.3 Usability and User Experience Metrics 34
4.5 Conclusion 34
Chapter 5: Experimental Results and Analysis 35
5.1 Introduction 36
5.2 AI Model Results 36
5.3 System Testing Results 37
5.4 Performance Evaluation 38
5.5 Conclusion 39
Chapter 6: Conclusion and Future Directions 40
6.1 Introduction 41
6.2 Conclusion 41
6.3 Future Direction 41
6.4 Conclusion 42
References 43

List of Tables

Table 1: Summary of Related Research Studies 8
Table 2: Performance of Mosquito Detection Model 36
Table 3: Performance of AI Chatbot Model 36
Table 4: Predictions of Mosquito Detection Model 37
Table 5: Summary of Test Case Results 37
Table 6: Detailed Test Execution Results 38
Table 7: System Performance Metrics 38
Table 8: Usability Evaluation 38

List of Figures

Figure 1: System Architecture Diagram 15
Figure 2: System Use-Case Diagram 16
Figure 3: System Sequence Diagram 20
Figure 4: System Component Diagram 21
Figure 5: Entity Relationship Diagram of the Database 22
Figure 6: Home Page Interface 23
Figure 7: News and Updates 23
Figure 8: Breeding Site Reporting Interface 24
Figure 9: AI Chatbot Interface 24
Figure 10: Mosquito Detection Interface 25
Figure 11: Admin Dashboard Interface 25

Abstract

Dengue fever remains a major public health concern in many tropical and subtropical regions, including Pakistan, where recurring outbreaks pose serious risks to communities and healthcare systems. Traditional dengue monitoring approaches often rely on delayed reporting and limited public awareness, which reduces the effectiveness of early prevention and response strategies. This project presents an **AI-Powered Dengue Monitoring and Mosquito Detection Platform** designed to assist in improving dengue awareness, monitoring potential risks, and supporting early identification of dengue-carrying mosquitoes. The platform integrates artificial intelligence techniques with a web-based information system to provide a centralized solution for dengue-related monitoring and public engagement. One of the key components of the system is a computer vision–based mosquito detection module that analyses images to determine whether the detected insect belongs to a mosquito class associated with dengue transmission. In addition to detection capabilities, the system provides an awareness module that delivers reliable information about dengue symptoms, prevention methods, and safety guidelines to users. The platform also allows users to report suspected mosquito breeding locations, enabling community participation in dengue surveillance and assisting authorities in identifying potential high-risk areas. The system architecture combines web technologies with machine learning models to process user inputs and generate detection results while maintaining an accessible and user-friendly interface. Through the integration of AI-based detection, monitoring mechanisms, and awareness resources, the platform demonstrates how technology can support dengue prevention initiatives by facilitating early identification, improving information accessibility, and encouraging community involvement in public health monitoring.

Chapter 1: Introduction

1.1 Introduction

Dengue fever is a rapidly spreading mosquito-borne disease that poses a significant public health challenge in many tropical and subtropical regions. Countries such as Pakistan experience recurring dengue outbreaks due to favourable environmental conditions and the presence of Aedes mosquitoes, which are responsible for transmitting the dengue virus. These outbreaks lead to increased health risks for communities and create additional pressure on healthcare systems. Effective dengue prevention requires timely monitoring, identification of mosquito breeding areas, and increased public awareness regarding preventive measures.

Traditional dengue monitoring approaches mainly rely on manual inspection and delayed reporting, which often limits the ability to respond quickly to potential outbreaks. In recent years, advancements in artificial intelligence and information technology have introduced new possibilities for improving disease monitoring and public awareness systems. Technologies such as machine learning and computer vision can analyze images and automatically detect objects, making it possible to identify mosquitoes through image-based analysis. Additionally, web-based platforms can be used to distribute reliable information and allow users to participate in reporting potential mosquito breeding sites.

This project focuses on the development of an AI-Powered Dengue Monitoring and Mosquito Detection Platform that integrates artificial intelligence with a web-based system to support dengue awareness and monitoring activities. The platform includes features for mosquito detection using image analysis, public awareness about dengue prevention, and community reporting of suspected mosquito breeding areas. By combining intelligent technologies with accessible information systems, the platform aims to contribute to improved awareness and monitoring related to dengue prevention.

1.2 Goal and Objectives

The primary goal of this project is to develop an intelligent platform that supports dengue monitoring and public awareness by utilizing artificial intelligence and web-based technologies. The system aims to provide users with tools that help identify mosquitoes, access reliable information about dengue prevention, and participate in reporting potential mosquito breeding areas.

The specific objectives of the project are as follows:

- To design and develop an **AI-based mosquito detection module** capable of identifying mosquitoes from images using computer vision techniques.
- To develop a **web-based platform** that provides users with information related to dengue symptoms, causes, and preventive measures.

- To implement a **community reporting feature** that allows users to report suspected mosquito breeding locations.
- To integrate artificial intelligence models with a web application framework to create a centralized dengue monitoring system.
- To provide an accessible and user-friendly interface that enables users to interact with the platform efficiently.
- To evaluate the functionality and performance of the developed system through testing and analysis.

1.3 Scope of the Project

The scope of this project is limited to the design and development of an **AI-Powered Dengue Monitoring and Mosquito Detection Platform** that supports dengue awareness and basic monitoring activities through a web-based system. The platform allows users to upload images for mosquito detection using an artificial intelligence–based model and provides informational resources related to dengue symptoms, causes, and preventive measures.

The system also includes a community reporting feature through which users can report suspected mosquito breeding locations. These reports help in identifying potential risk areas and encourage community participation in dengue prevention efforts. The platform is designed to be accessible through a web interface so that users can easily interact with the system and utilize its features.

However, the system is not intended to function as an official medical diagnostic tool or government surveillance system. Its purpose is to assist in awareness, provide basic monitoring support, and demonstrate the use of artificial intelligence in addressing public health challenges related to dengue. The project scope is therefore limited to the development, integration, and evaluation of the platform's main modules, including mosquito detection, awareness information, and community reporting.

1.4 Conclusion

In this chapter, the problem of dengue monitoring and the need for improved solutions were discussed. The objectives and scope of the proposed system was defined, providing a clear direction for the project. This forms the foundation for understanding the system and its development in the following chapter

Chapter 2: Literature Review

2.1 Introduction

Dengue fever is a mosquito-borne disease that poses a significant public health challenge in many tropical regions, including Pakistan. Traditional dengue monitoring methods rely mainly on manual surveillance and delayed reporting, which limits early detection and prevention.

Recent research has explored the use of artificial intelligence, machine learning, and computer vision techniques to improve dengue monitoring and mosquito detection methods[1], [2]. These technologies can help identify dengue risks more efficiently and support prevention efforts.

This chapter reviews existing studies related to dengue prediction, mosquito detection, and AI-based monitoring systems to understand current approaches and identify research gaps addressed by the proposed system.

2.2 Background and Problem Elaboration

Dengue fever is transmitted primarily through the bite of infected *Aedes aegypti* mosquitoes. These mosquitoes breed in stagnant water and are commonly found in urban environments where water containers, drains, and other water-holding structures exist. The rapid spread of dengue has become a serious public health issue in many countries.

In Pakistan, dengue outbreaks have occurred repeatedly over the past decade, particularly in major cities such as **Lahore**, **Islamabad**, and **Karachi**. These outbreaks highlight the need for improved monitoring systems and increased public awareness about dengue prevention.

Traditional dengue monitoring systems often depend on manual inspections and reporting by health authorities. Such approaches can be time-consuming and may not detect potential outbreaks early enough. As a result, researchers have started investigating the use of artificial intelligence, machine learning, and computer vision techniques to support dengue surveillance and mosquito detection.

By using intelligent technologies, it is possible to analyze large datasets, identify mosquito species from images, and develop systems that assist communities and authorities in monitoring dengue risks more efficiently.

2.3 Detailed Literature Review

2.3.1 Definitions

- **Dengue Fever**

  Dengue fever is a viral infection transmitted by *Aedes* mosquitoes, primarily *Aedes aegypti*. It causes symptoms such as fever, headache, muscle pain, and in severe cases can lead to dengue hemorrhagic fever.

- **Machine Learning**

  Machine learning is a subset of artificial intelligence that allows computer systems to learn patterns from data and make predictions or decisions without being explicitly programmed.
- **Computer Vision**

  Computer vision is a field of artificial intelligence that enables computers to interpret and analyze visual information such as images and videos.
- **Mosquito Detection**

  Mosquito detection refers to the use of image processing or machine learning techniques to identify mosquito species or larvae from visual data.

2.3.2 Related Research Work 1

Hussain et al. (2023) reviewed different machine learning techniques used for predicting dengue outbreaks[1]. The study analyzed various algorithms such as Support Vector Machines, Decision Trees, and Neural Networks to evaluate their effectiveness in predicting dengue spread based on environmental and epidemiological data[1]. The results indicated that machine learning models can analyze climate variables such as rainfall, humidity, and temperature to identify patterns related to dengue outbreaks. The research highlighted the importance of intelligent systems in improving dengue surveillance and early warning systems.

2.3.3 Related Research Work 2

Another study investigated the use of Convolutional Neural Networks (CNN) for detecting mosquito larvae from images. The research demonstrated that deep learning models can automatically detect mosquito larvae with high accuracy by analyzing images of water containers where mosquitoes breed[2]. The findings showed that computer vision techniques can significantly reduce the need for manual inspection in mosquito monitoring. However, the system mainly focused on larvae detection in controlled environments and did not integrate the detection mechanism into a public monitoring platform.

2.3.4 Related Research Work 3

A separate research study explored the use of machine learning models for predicting dengue outbreaks using environmental and epidemiological data. The researchers analyzed climate variables such as temperature, rainfall, and humidity to develop predictive models capable of identifying high-risk dengue areas. The findings showed that machine learning can improve early warning systems and help authorities plan preventive measures[3]. However, the study primarily focused on prediction

using historical data and did not include real-time mosquito detection or public participation features.

2.4 Literature Review Summary Table

Table 1: Summary of Related Research Studies

The summary of selected research studies related to dengue monitoring, mosquito detection, and machine learning approaches is presented here.

No	Name	Inventor	Year	Input	Output	Description
1	Machine Learning Approaches for Dengue Prediction [1]	Zafer Hussain et al.	2023	Environmental data, dengue case records	Dengue outbreak prediction	The study reviews multiple machine learning algorithms such as Decision Trees, Support Vector Machines, and Neural Networks to predict dengue outbreaks using climatic and epidemiological data.
2	Detection of Mosquito Larvae Using CNN [2]	Meer Shadman Saeed et al.	2021	Images of water containers	Presence or absence of mosquito larvae	A deep learning model based on Convolutional Neural Networks is used to detect mosquito larvae from images, helping automate mosquito surveillance and mosquito control.
3	Data-Driven Machine Learning Models for Predicting Dengue Outbreak [3]	Bushra Mazhar et al.	2024	Climate data and dengue case data	Dengue outbreak risk prediction	The research analyzes climatic variables such as rainfall, temperature, and humidity using machine learning techniques to forecast dengue outbreaks and support early warning systems.
4	Deep Learning Approach for Identifying Dengue Mosquito Larvae [4]	Ali Saeed et al.	2022	Mosquito larvae image dataset	Identification of dengue mosquito larvae	The study uses deep learning and image classification techniques to identify dengue mosquito larvae from images, improving mosquito monitoring and supporting automated vector surveillance systems.

2.5 Research Gap

Although several studies have explored machine learning techniques for dengue prediction and deep learning methods for mosquito detection, most of these systems focus on a single aspect of dengue monitoring[1], [2], [3], [4]. Some research focuses on predicting dengue outbreaks using environmental data, while others focus on detecting mosquito larvae using image processing techniques.

However, there is a lack of integrated systems that combine **AI-based mosquito detection, public**

awareness, and community reporting features in a single platform. Additionally, many existing systems are designed mainly for research purposes and are not easily accessible to the general public. Therefore, there is a need for a system that integrates artificial intelligence with a user-friendly platform to support dengue monitoring, awareness, and community participation.

2.6 Problem Statement

Dengue fever continues to be a major public health concern, particularly in regions where mosquito populations are high and awareness about dengue prevention is limited. Traditional monitoring systems rely heavily on manual inspections and delayed reporting, which makes it difficult to detect potential outbreaks early.

The problem addressed in this project is the lack of accessible and intelligent platforms that assist in mosquito detection and dengue awareness. To address this issue, this project proposes the development of an **AI-Powered Dengue Monitoring and Mosquito Detection Platform** that utilizes artificial intelligence and web technologies to support dengue monitoring and awareness activities. The system aims to provide users with tools for mosquito detection, educational resources about dengue prevention, and the ability to report potential mosquito breeding sites.

2.7 Conclusion

The literature review highlighted various techniques used for dengue prediction and mosquito detection. While existing studies provide useful insights, they often focus on individual aspects of the problem. This creates an opportunity for developing an integrated system that combines detection, reporting, and awareness features.

Chapter 3: Requirements and Design

3.1 Introduction

This chapter describes the system requirements and overall design of the proposed **AI-Powered Dengue Monitoring and Mosquito Detection Platform**. The purpose of this chapter is to explain the functional and non-functional requirements needed for the development of the system, as well as the hardware and software resources required for implementation. It also presents the methodology used to develop the system and the overall system architecture. The information provided in this chapter allows readers to understand how the system is designed and how its components interact to achieve the objectives of the project.

3.2 Requirements

The requirements of a system define the functionalities, constraints, and resources needed for the successful development and operation of the proposed platform. For the AI-Powered Dengue Monitoring and Mosquito Detection Platform, the requirements include functional requirements, non-functional requirements, and the hardware and software resources necessary to develop and run the system. These requirements ensure that the system performs efficiently, meets user needs, and operates reliably in real-world environments. By defining these requirements clearly, the development process becomes more structured and helps ensure that the final system achieves its intended goals.

3.2.1 Functional Requirements

Functional requirements describe the core operations and services that the system must perform. These requirements define how the system interacts with users and how different modules of the platform operate to achieve the objectives of dengue monitoring and awareness.

The proposed system includes the following functional requirements:

1. User Registration and Authentication

The system should allow users to create accounts and securely log in to access the platform. This ensures that user activities such as reporting breeding sites or using AI detection features are properly associated with user profiles.

2. Mosquito Detection Module

The system should allow users to upload images of mosquitoes or mosquito breeding areas. The AI model integrated into the system should analyze the uploaded images and determine whether the detected object is a mosquito or not. This feature helps users identify potential dengue-carrying mosquitoes.

3. Breeding Site Reporting Module

Users should be able to report potential mosquito breeding sites through the platform. The system should allow users to submit information such as location, description, and images of the breeding site. These reports can help authorities and communities identify high-risk areas.

4. Dengue Awareness and Information Module

The platform should provide educational content about dengue fever, including prevention tips, symptoms, and methods to reduce mosquito breeding. This module aims to increase public awareness and promote preventive measures.

5. Dashboard and Monitoring Module

The system should provide an administrative dashboard where data related to mosquito detection results, reported breeding sites, and system usage statistics can be monitored. This feature helps administrators analyze trends and manage reports submitted by users.

6. Data Storage and Management

The system should store user data, image data, and report information in a database. The stored data can be used for system analysis, monitoring dengue risks, and improving AI model performance.

7. News and Updates Module

The system provides a section where users can view the latest dengue-related news and updates. This content is managed manually by the administrator.

3.2.2 Hardware and Software Requirements

This section describes the hardware and software components required to develop and run the proposed system.

Hardware Requirements

- Computer system with at least **8 GB RAM**
- Minimum **Intel Core i5 processor or equivalent**
- Stable internet connection
- GPU (optional but recommended) for training AI models

Software Requirements

- **Programming Languages:** Python, JavaScript
- **Frameworks/Libraries:** TensorFlow / PyTorch, React / Next.js, Node.js
- **Database:** SQlite
- **Development Tools:** VS Code, Git, GitHub. Kaggle
- **Operating System:** Windows

These tools and resources are required to develop the AI models, build the web platform, and deploy

the system effectively.

3.2.3 Proposed Methodology

The proposed methodology describes the approach used to design, develop, and implement the AI-Powered Dengue Monitoring and Mosquito Detection Platform. The system combines artificial intelligence techniques with a web-based platform to provide a comprehensive solution for dengue monitoring and awareness.

The development process begins with **data collection and preparation**, where image datasets containing mosquito images and non-mosquito images are gathered. These datasets are then preprocessed to improve the quality of the training data by resizing images, normalizing values, and organizing the dataset into training and testing sets.

Next, a **deep learning model** based on Convolutional Neural Networks (CNN) is trained to detect mosquito images. The trained model learns to recognize visual patterns associated with mosquitoes and differentiate them from other objects[2], [4]. Once the training process is completed, the model is evaluated to measure its accuracy and performance[1], [3].

After the AI model is developed, it is **integrated into the web platform** so that users can upload images and receive detection results. The backend server processes the uploaded images, sends them to the AI model for analysis, and returns the prediction results to the user interface.

In addition to mosquito detection, the system also includes modules for **reporting breeding sites and providing dengue awareness information**. These modules allow users to actively participate in monitoring mosquito activity and help communities identify high-risk areas.

Finally, the platform includes a **data management and monitoring system** that stores reports and detection results in a database. This data can be analyzed to identify patterns and trends related to mosquito presence and potential dengue risks.

The overall methodology ensures that the system integrates artificial intelligence with community participation to create an effective and accessible dengue monitoring solution.

3.3 System Architecture

The system architecture describes the overall structure of the proposed **AI-Powered Dengue Monitoring and Mosquito Detection Platform** and explains how different components of the system interact with each other. The architecture integrates a web-based platform with artificial intelligence modules to provide mosquito detection, breeding site reporting, and dengue awareness features.

The system follows a modular architecture in which the frontend interface, backend services, database, and AI model operate as interconnected components. The frontend handles user interaction and communicates with the backend through API requests. The backend processes user requests, manages data, and interacts with the AI model for mosquito detection. The AI module analyzes uploaded images using a trained deep learning model and returns prediction results[2], [4]. All system data such as user information, detection results, and reported breeding sites are stored in a centralized database.

The following subsections describe the major architectural elements of the system including the system architecture diagram, component design, data flow, database structure, and AI model pipeline.

3.3.1 System Architecture Diagram

The system architecture diagram provides a high-level overview of the structure of the proposed platform. It illustrates how the main components of the system interact with each other to perform different tasks. The system consists of three primary layers: the presentation layer, application layer, and data layer.

The **presentation layer** represents the user interface through which users interact with the platform. This layer allows users to register, log in, upload mosquito images for detection, report breeding sites, and access dengue awareness information.

The **application layer** contains the backend services and the AI processing module. The backend server handles requests from the user interface, processes user inputs, communicates with the database, and sends images to the AI detection model for analysis. The AI module uses a trained deep learning model to classify whether the uploaded image contains a mosquito.

The **data layer** includes the database system that stores user information, detection results, reports of mosquito breeding sites, and other system data. This layer ensures that all relevant information is stored securely and can be retrieved when required.

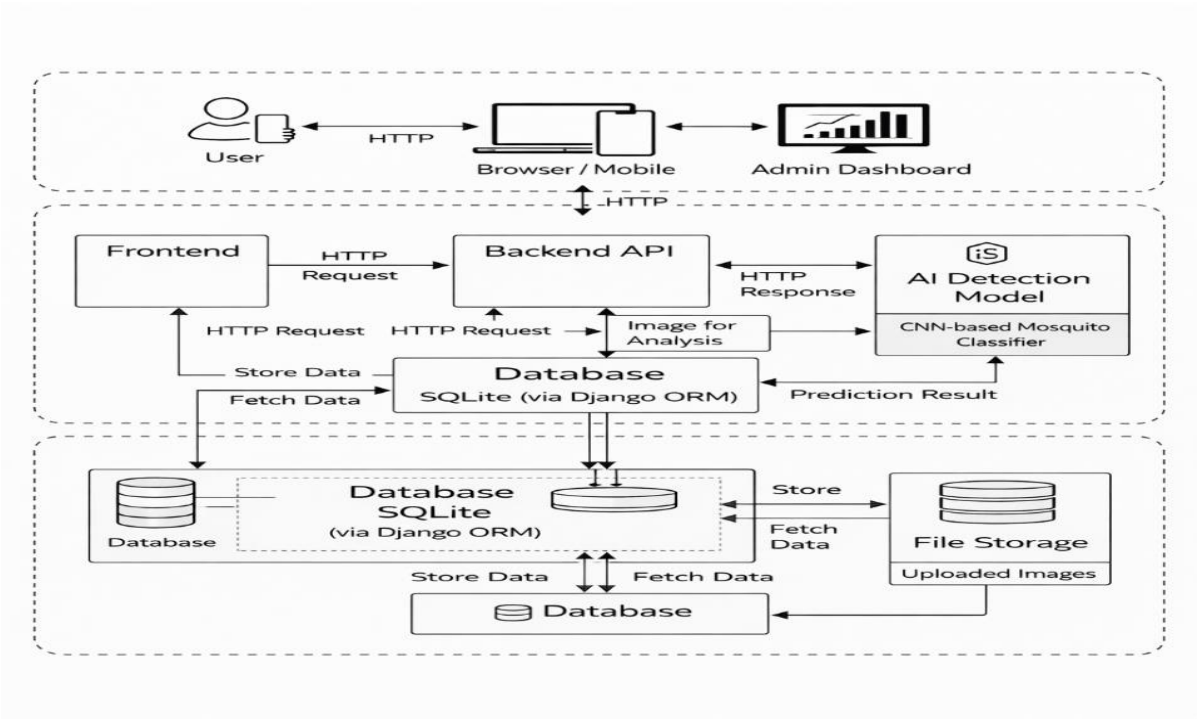

Figure 1: System Architecture Diagram

This figure shows the overall system architecture, including frontend, backend, AI module, and database, and how they interact to process user requests.

3.4 Use Cases

This section describes the major interactions between users and the proposed dengue monitoring platform. Use cases illustrate how different actors interact with the system to perform specific tasks. These interactions help define the behavior of the system and explain how system features are used in real scenarios. The primary actors involved in the system include registered users and system.

administrators. Users can perform activities such as uploading mosquito images for detection, reporting potential breeding sites, and accessing dengue awareness information. Administrators are responsible for monitoring system reports and managing platform data.

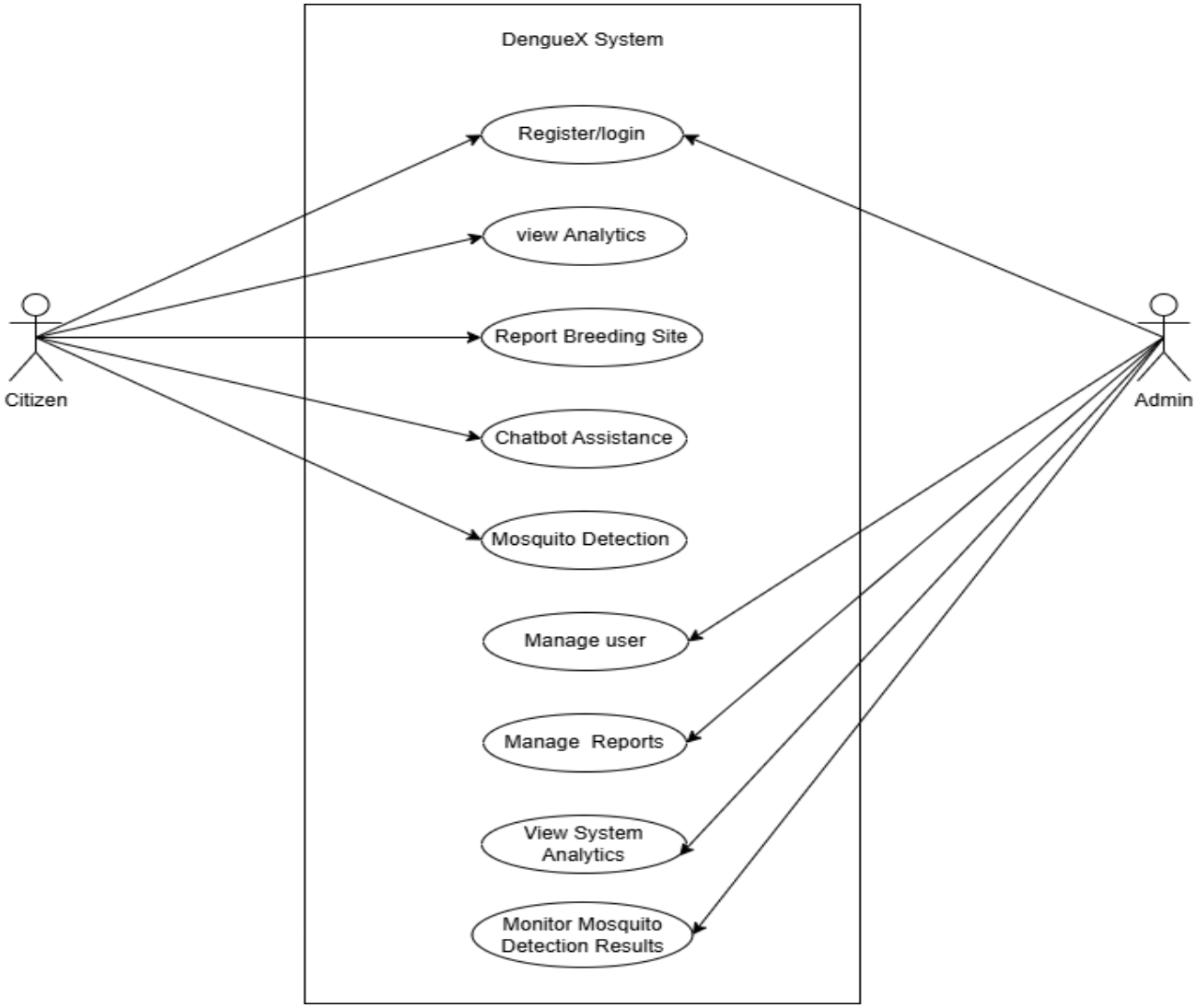

Figure 2: System Use-Case Diagram

This figure illustrates the use case diagram of the system, showing the interactions between different actors and the system's main functionalities.

3.4.1 User Authentication

Name	User Authentication
Actors	Registered User, Administrator
Summary	The user provides login credentials to access the dengue monitoring platform. After successful verification, the system allows the user to access the dashboard and available system features.
Pre-Conditions	User must have a registered account
Post-Conditions	User is successfully logged in and redirected to dashboard

Special Requirements	Internet connection and valid login credentials are required		
Basic Flow			
Actor Action		**System Response**	
1	The user navigates to the login page.	2	The system displays the login interface requesting email and password.
3	The user enters valid login credentials.	4	The system verifies the credentials, creates a user session, and redirects the user to the dashboard.
Alternative Flow			
3	The user enters an incorrect email or password.	4-A	The system displays an error message indicating that the login information is incorrect and requests the user to try again.

3.4.2 Mosquito Image Detection

Name	Mosquito Image Detection		
Actors	Registered User		
Summary	The user uploads an image to the platform, and the system analyzes the image using the AI detection model to determine whether it contains a mosquito.		
Pre-Conditions	The user must be logged into the system and have access to the image upload feature.		
Post-Conditions	The system generates a prediction result and displays it to the user. The detection result may also be stored in the system database.		
Special Requirements	The uploaded image must be in a supported format such as JPG or PNG.		
Basic Flow			
Actor Action		**System Response**	
1	The user selects the mosquito detection feature.	2	The system displays an image upload interface.
3	The user uploads an image for analysis.	4	The system processes the image using the AI model and generates a detection result.
Alternative Flow			
3	The user uploads an unsupported file format.	4-A	The system displays a message requesting a valid image file.

3.4.3 Breeding Site Reporting

Name	Breeding Site Reporting		
Actors	Registered User		
Summary	Users can report suspected mosquito breeding sites by submitting location details, description, and optional images.		
Pre-Conditions	The user must be logged into the platform.		
Post-Conditions	The report is saved in the database and becomes available for administrative review.		
Special Requirements	The system should allow location input or image attachment for the report.		
Basic Flow			
Actor Action		**System Response**	
1	The user selects the report breeding site option.	2	The system displays a reporting form.
3	The user enters location details and description.	4	The system saves the report in the database.
		5	The system confirms successful submission.
Alternative Flow			
3	The user submits incomplete information.	4-A	The system displays a message requesting required information before submission.

3.4.4 Analytics Dashboard

Name	Analytics Dashboard Interaction		
Actors	User		
Summary	The user accesses the analytics dashboard to view visual insights related to mosquito detections and reported breeding sites. The dashboard presents statistical information through charts and graphs to help users understand dengue risk patterns.		
Pre-Conditions	The user must be logged into the system and have access to the dashboard interface.		
Post-Conditions	Dashboard data is displayed successfully.		
Special Requirements	The system must have stored data available for visualization.		
Basic Flow			
Actor Action		**System Response**	

1	The user opens the dashboard	2	The system displays analytics (charts, KPIs, map).
3	The user hovers on the map.	4	System shows location-based data
Alternative Flow			
3	No data available	4-A	System displays “No data available”

3.4.5 AI Chatbot

Name	AI Chatbot Interaction		
Actors	Registered User		
Summary	The user interacts with the AI chatbot to receive information related to dengue prevention, symptoms, and general guidance. The chatbot provides responses based on user queries through a conversational interface.		
Pre-Conditions	The user must be logged into the system and have access to the chatbot interface.		
Post-Conditions	System provides response to user query		
Special Requirements	The system generates and displays an appropriate response to the user’s query.		
Basic Flow			
Actor Action		**System Response**	
1	User opens chatbot interface	2	System displays chat window
3	User enters a query	4	System processes the query
5	System generates response	6	Response is displayed to user
Alternative Flow			
3	The user enters an unclear or unsupported query.	4-A	System asks user to rephrase the query

3.5 Sequence Diagram

The sequence diagram illustrates the interaction between different components of the proposed dengue monitoring system during mosquito detection and report management. The process begins when the user uploads a mosquito image through the system interface. The system forwards the image to the AI detection model for analysis. The AI model processes the image and generates a prediction result indicating whether the uploaded image contains a mosquito.

Once the prediction is generated, the system stores the detection result in the database and returns the result to the user interface, where it is displayed to the user. In addition to mosquito detection, users can

submit breeding site reports through the platform. These reports are stored in the database and can be accessed by administrators through the admin dashboard for monitoring and analysis. The sequence diagram demonstrates the order of interactions between the user, system, AI model, database, and administrator during these operations.

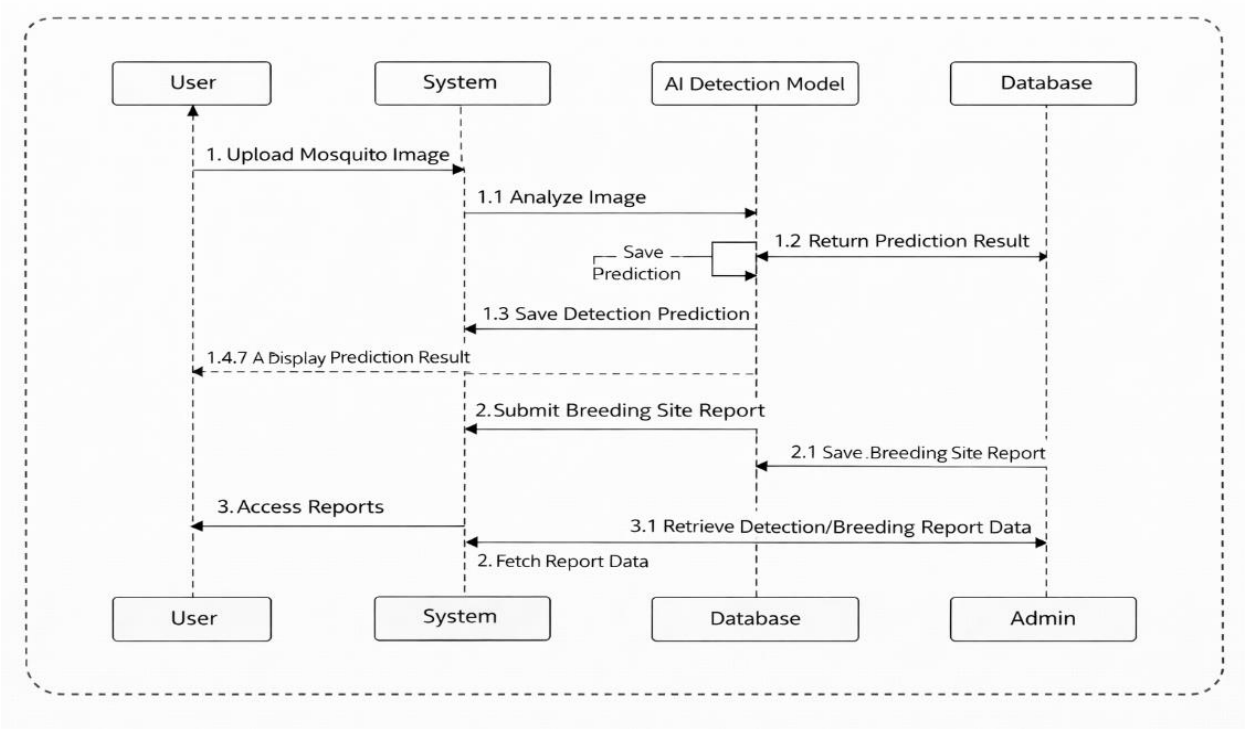

Figure 3: System Sequence Diagram

This diagram shows the interaction flow between the user, system, AI model, and database during detection and reporting processes.

3.6 Component Diagram

The component diagram represents the internal modules of the system and their relationships. Each component of the system performs a specific function and interacts with other components to achieve the overall functionality of the platform.

The major components of the system include the **User Interface Module**, **Authentication Module**, **Mosquito Detection Module**, **Breeding Site Reporting Module**, **Awareness Information Module**, **Database Module**, and **Administrative Dashboard**.

The **User Interface Module** is responsible for displaying system features to users and collecting user inputs such as image uploads or report submissions. The **Authentication Module** manages user registration and login processes to ensure that only authorized users can access certain system features.

The **Mosquito Detection Module** integrates the trained deep learning model used for image classification. When a user uploads an image, this module processes the image and generates prediction results indicating whether the image contains a mosquito.

The **Breeding Site Reporting Module** allows users to submit reports about potential mosquito breeding areas. These reports are stored in the database and can be reviewed by administrators.

The **Awareness Information Module** provides educational content related to dengue prevention, symptoms, and mosquito control practices. Finally, the **Administrative Dashboard** enables system administrators to monitor reports, analyze data, and manage the platform.

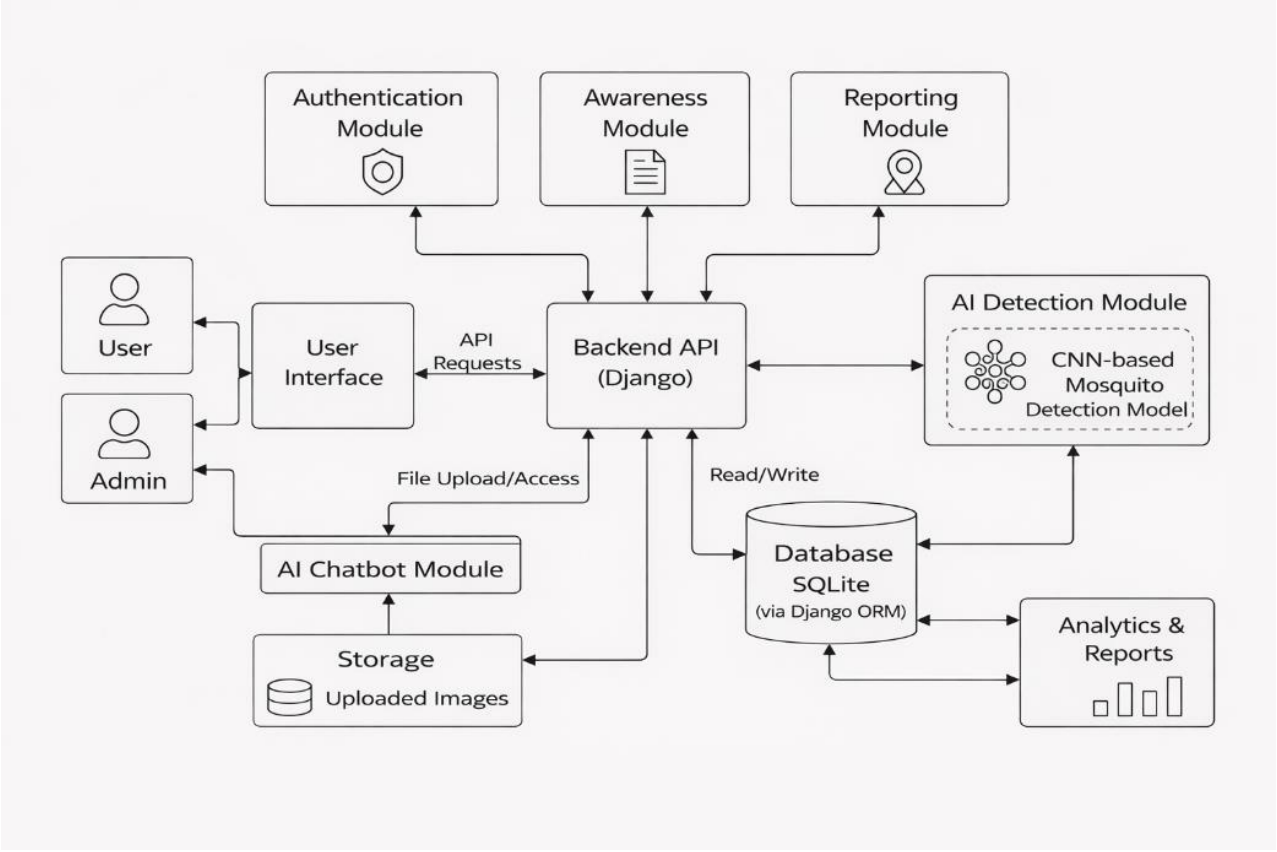

Figure 4: System Component Diagram

This figure presents the main system components and their relationships, including UI, backend services, AI detection module, and database.

3.7 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) describes the structure of the database used in the proposed system. It identifies the main entities in the database and the relationships between them.

The primary entity in the system is the **User** entity, which stores information about registered users such

as user ID, name, email, and login credentials. Each user can perform multiple actions within the platform, such as uploading mosquito images for detection or reporting breeding sites.

Another important entity is the **Mosquito Detection** entity, which stores information related to uploaded images and prediction results generated by the AI model. Each detection record is associated with a specific user who uploaded the image.

The **Breeding Site Report** entity stores details about reported mosquito breeding areas, including location information, description, and images provided by users. These reports help identify high-risk areas where mosquitoes may breed.

Additional entities such as **Awareness Content** or **Administrator** may also be included to manage educational information and system operations. The relationships between these entities ensure that user activities and system data are properly organized and maintained within the database.

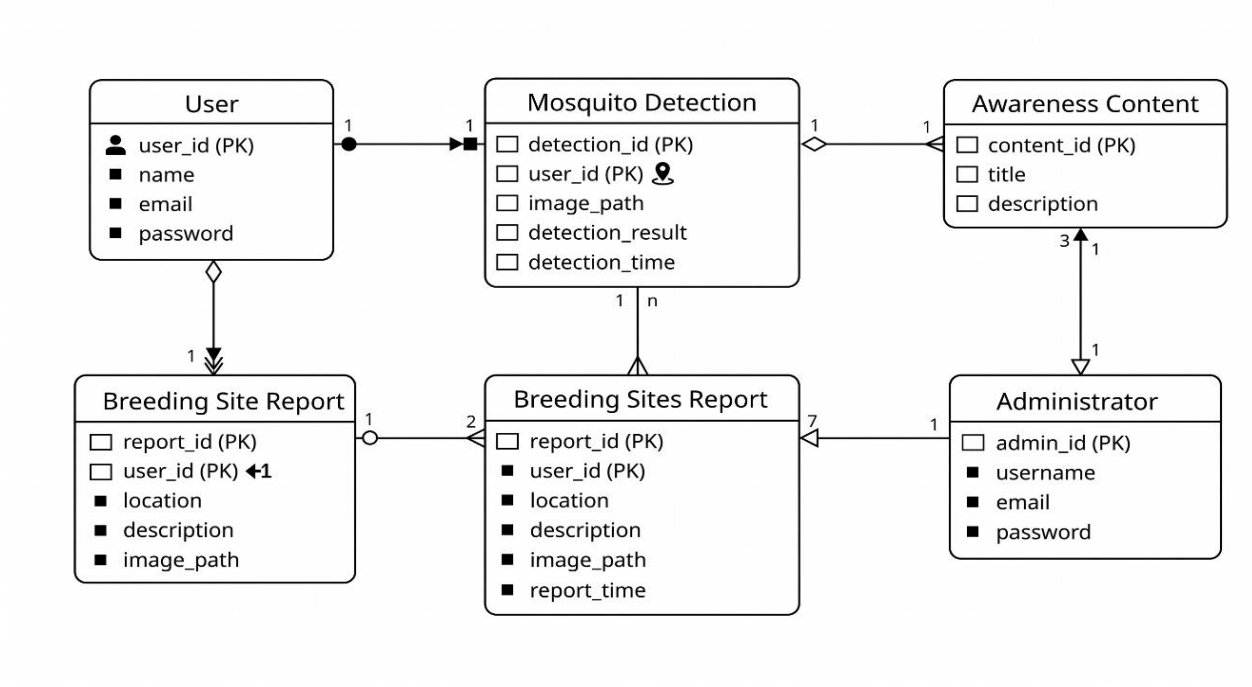

Figure 5: Entity Relationship Diagram of the Database

This figure shows the database structure and relationships between entities such as User, Detection, Reports, and Administrator.

3.8 Graphical User Interfaces

This section presents the graphical user interface (GUI) of the proposed AI-Powered Dengue Monitoring and Mosquito Detection Platform. The interface is designed to be user-friendly, visually appealing, and easy to navigate. It allows users to interact with different modules of the system, including detection,

reporting, analytics, and chatbot assistance.

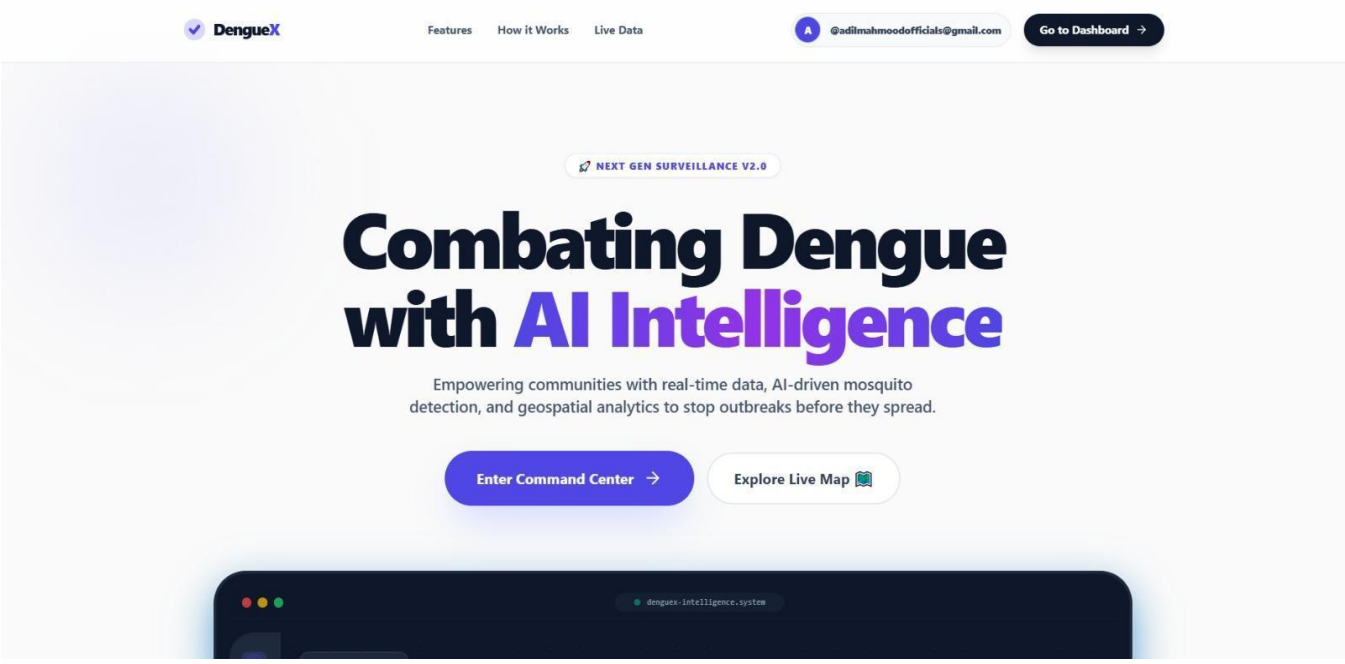

Figure 6: Home Page Interface

This interface represents the landing page of the system, providing an overview of the platform's purpose and features. It allows users to navigate to key sections such as the dashboard and live monitoring features.

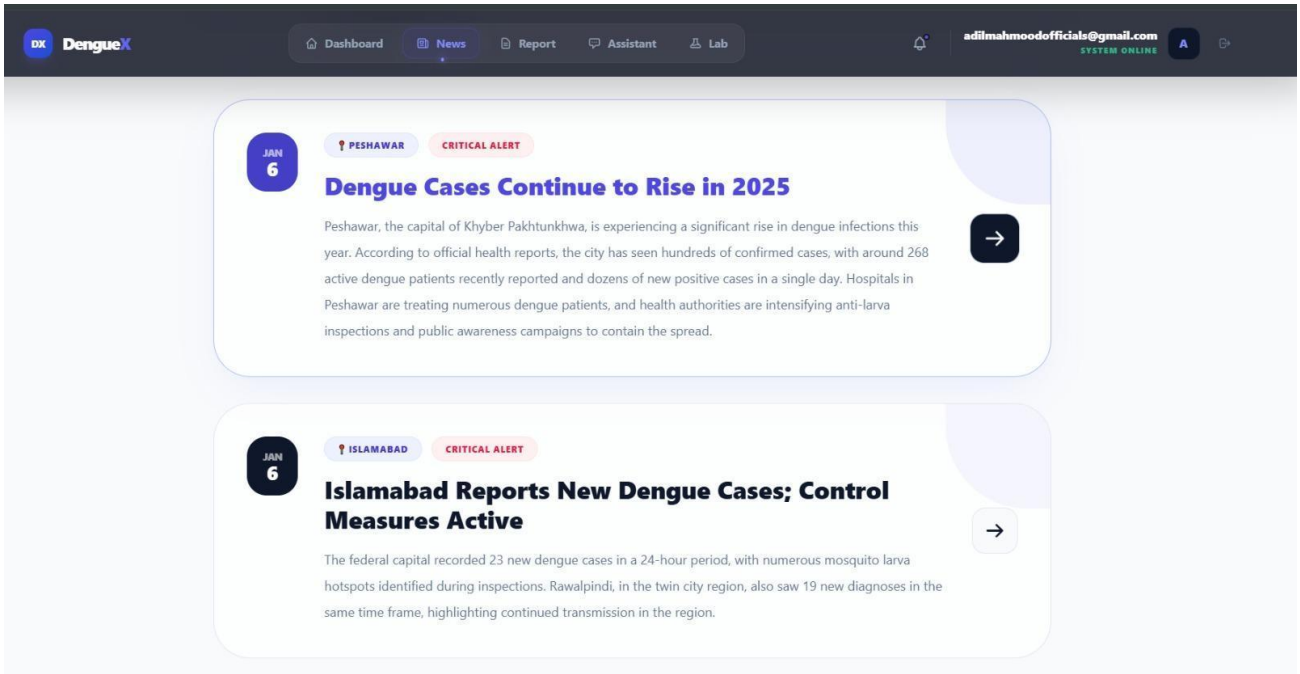

Figure 7: News and Updates

This screen displays dengue-related news and alerts managed by the administrator. It helps users stay informed

about recent dengue cases and preventive measures.

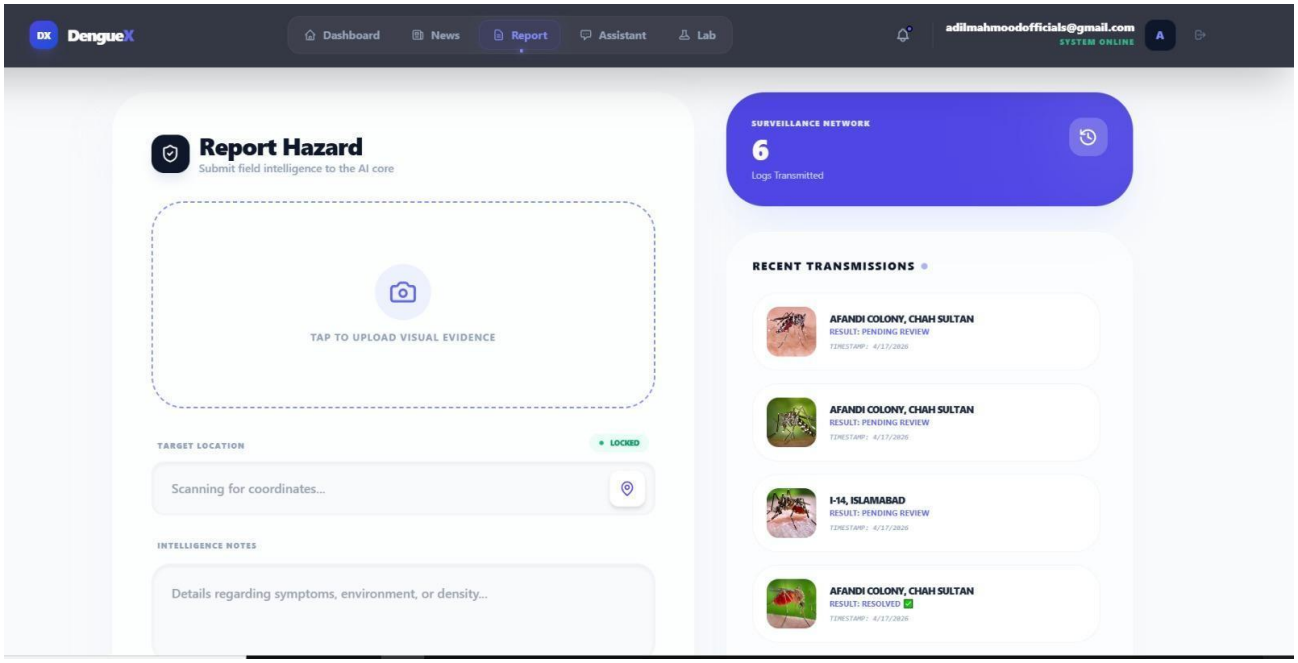

Figure 8: Breeding Site Reporting Interface

This interface allows users to report mosquito breeding sites by uploading images and providing location details. It supports community participation in dengue monitoring.

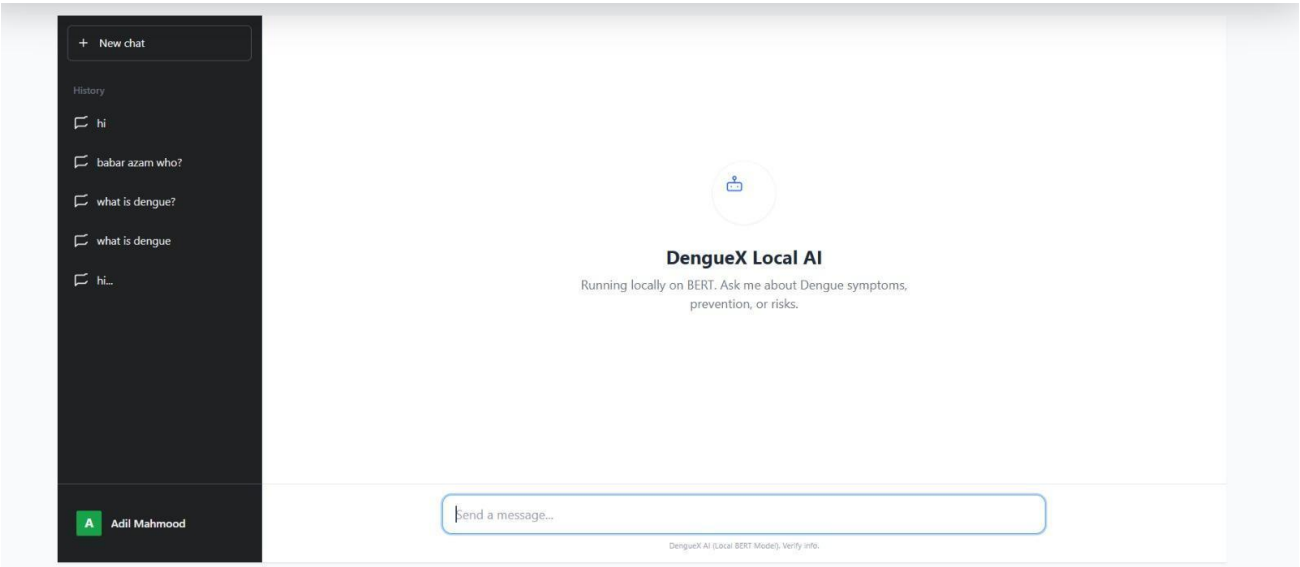

Figure 9: AI Chatbot Interface

This screen shows the chatbot interface, where users can ask questions related to dengue symptoms, prevention, and safety guidelines. The chatbot provides instant responses to user queries.

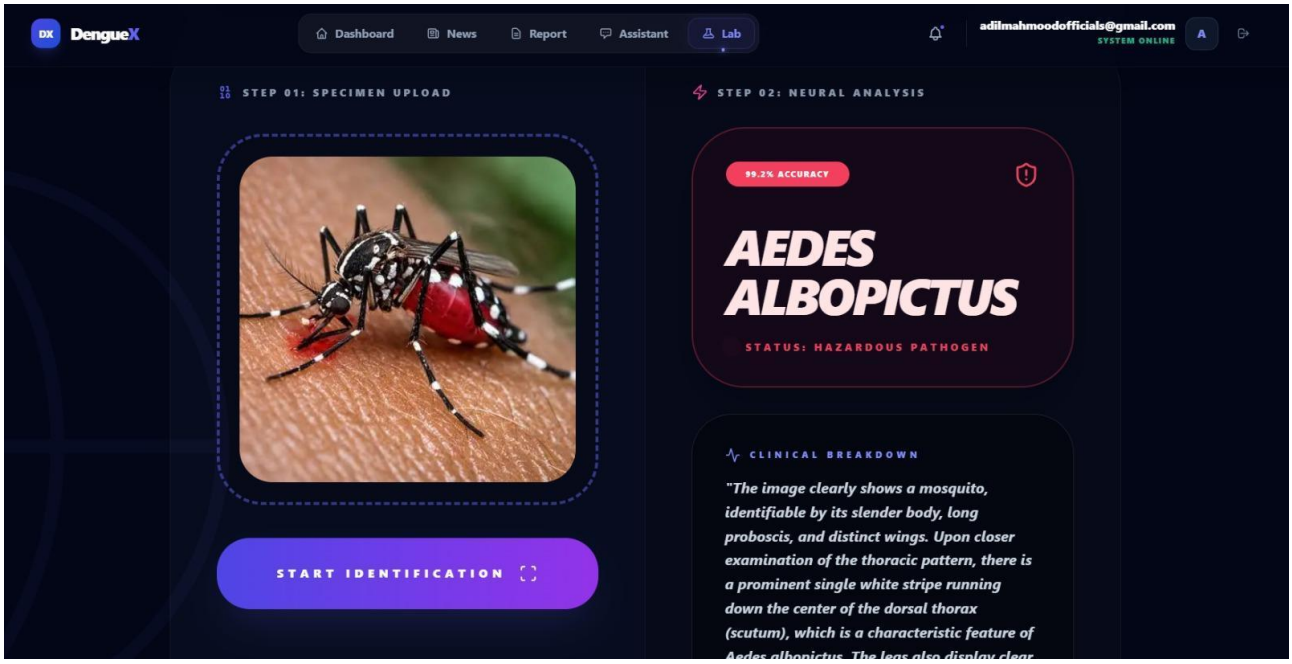

Figure 10: Mosquito Detection Interface

This interface displays the mosquito detection module, where users upload images for analysis. The system processes the image using the AI model and presents the detection results.

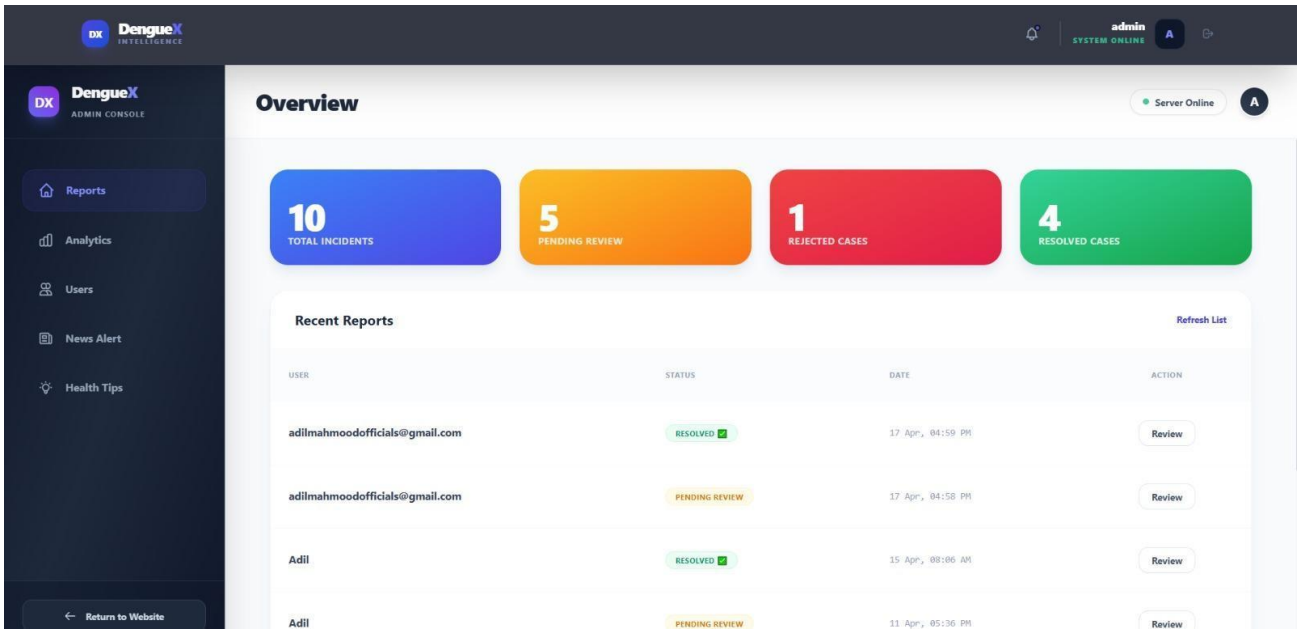

Figure 11: Admin Dashboard Interface

This screen represents the administrative dashboard, where administrators can monitor reports, manage data, and analyze system activities through visual summary

Chapter 4: Implementation and Test Cases

4.1 Introduction

This chapter describes the implementation details and testing procedures of the proposed **AI-Powered Dengue Monitoring and Mosquito Detection Platform**. It explains how the system components were developed, including the AI-based mosquito detection module and the web application. In addition, this chapter presents the test case design and evaluation metrics used to verify system functionality and performance.

4.2 Implementation

The system was implemented using a combination of web development technologies and artificial intelligence techniques. The backend was developed using the **Django framework**, which handles server-side logic, API processing, and database operations. The frontend provides a user-friendly interface for interacting with the system.

The system includes several modules such as user authentication, mosquito detection, breeding site reporting, and an analytics dashboard. Data generated by these modules is stored using **SQLite**, which is integrated with Django through its ORM.

The mosquito detection module was implemented using a **Convolutional Neural Network (CNN)** model. The model was trained on image datasets containing mosquito and non-mosquito samples. CNN-based methods are commonly used for mosquito and larvae detection tasks in previous studies[2], [4].

The system also supports community participation through a reporting feature that allows users to submit breeding site information. These reports are stored in the database and can be analyzed through the dashboard, supporting dengue monitoring activities.

4.2.1 Implementation of Mosquito Detection Module

The main component of the system is the mosquito detection module. The implementation process started with collecting image data containing mosquito and non-mosquito samples. The dataset was preprocessed by resizing images and normalizing pixel values.

A Convolutional Neural Network (CNN) was then trained using the prepared dataset. The model learns distinguishing features such as shape and texture to identify mosquitoes. Similar approaches have been successfully used in related research[2], [4]

After training, the model was tested using validation data to evaluate its accuracy. Once acceptable performance was achieved, the model was integrated into the backend system. When a user uploads an image, the system processes it through the model and returns a prediction result.

4.2.2 Implementation of User Authentication Module

The user authentication module was implemented using Django's built-in authentication framework, which provides a secure and scalable mechanism for managing user access. The system supports user registration, login, logout, and session handling. Passwords are securely stored using hashing techniques, ensuring protection against unauthorized access.

Additionally, session-based authentication is utilized to maintain user state across different pages. This approach ensures that only authenticated users can access restricted functionalities such as reporting breeding sites or viewing analytics. Security practices such as input validation and protection against common vulnerabilities (e.g., SQL injection and cross-site scripting) were also considered during implementation.

Although this module is not directly related to dengue prediction, secure data handling is critical in health-related systems where user-submitted data contributes to surveillance and analysis processes[1].

4.2.3 Implementation of Breeding Site Reporting Module

The breeding site reporting module enables users to actively participate in dengue prevention by reporting mosquito breeding locations. A structured web form was designed to collect relevant data such as geographic location, environmental description, and optional image uploads.

The collected data is stored in a centralized database using Django ORM, ensuring efficient data management and retrieval. This module aligns with real-world dengue surveillance practices, where environmental and location-based data are essential for identifying high-risk areas.

Environmental factors such as stagnant water, rainfall, and humidity significantly influence mosquito breeding and disease spread[4].

In addition, mosquito detection research highlights the importance of identifying breeding sites to control vector populations effectively. Community-based reporting further enhances surveillance systems and supports early intervention strategies[3].

4.2.4 Implementation of Analytics Dashboard

The analytics dashboard was developed to visualize system data and support decision-making. It displays trends related to reported breeding sites and dengue-related indicators using charts and summaries.

The dashboard processes stored data and presents it in graphical formats, helping users and administrators identify patterns over time. This is consistent with data-driven approaches discussed

in[4], where statistical analysis and visualization techniques are used to understand correlations between climatic variables and dengue incidence.

For example, correlation techniques such as Pearson correlation can be used to analyze relationships between variables like humidity, temperature, and dengue cases. Furthermore, machine learning approaches use regression and time-series models to predict future outbreaks based on historical data[3], [4].

This module provides a simplified version of such analytical insights for practical system use.

4.2.5 Implementation of AI Chatbot Module

The AI chatbot module was implemented to provide users with instant dengue-related guidance through a conversational interface. The chatbot processes user queries and generates responses based on predefined datasets and rule-based logic.

While the current implementation uses basic natural language processing techniques, it reflects the broader role of artificial intelligence in healthcare systems. Deep learning and AI-based models have shown strong potential in analyzing epidemiological and environmental data for disease prediction and monitoring[1], [4].

Machine learning approaches further demonstrate how intelligent systems can assist in decision-making and awareness generation in healthcare applications. The chatbot enhances user engagement and supports public awareness by providing accessible and timely information.

4.2.6 Implementation of News and Updates Module

The news and updates module was designed to provide users with the latest information related to dengue outbreaks and prevention measures. This module allows administrators to manually add and manage content, which is then displayed on the platform.

Although simple in functionality, this module plays a crucial role in awareness dissemination. Public awareness and timely information are essential components of dengue control strategies, alongside predictive modelling and surveillance systems[3], [4].

By integrating updated information with system data, this module helps bridge the gap between data analysis and real-world preventive action.

4.3 Test Case Design and Description

This section describes the test cases developed to verify system functionality. The test cases cover all major modules including authentication, mosquito detection, reporting, analytics, and administration. Each test case includes input conditions, execution steps, and expected results.

These tests ensure that the system behaves correctly under normal and error conditions.

4.3.1 User Authentication Module Test Case

User Authentication Module			
Authentication Case			
Test Case ID:	TC-01	**Test Date:**	19/04/2026
Test case Version:	1.0	**Use Case Reference(s):**	User Authentication
Revision History:	v1.0 – Basic authentication implemented in FYP-I v2.0 – Test case formally documented and structured in FYP-II v2.1 – Minor refinements in validation and login flow		
Objective	To verify that the user can log in with valid credentials		
Product/Ver/Module:	Dengue Monitoring System – Authentication Module		
Environment:	Web browser, Django backend		
Assumptions:	The user is already registered.		
Pre-Requisite:	User account exists in database		

Step No.	**Execution description**	**Procedure result**
1	User opens login page	Login interface is displayed
2	User enters valid email and password	System accepts input
3	User clicks login button	System verifies and redirects to dashboard
Comments: Login functionality works as expected		
☑ Passed ☐ Failed ☐ Not Executed		

4.3.2 Mosquito Detection Module Test Case

Mosquito Detection Module			
Detection Use Case			
Test Case ID:	TC-02	**Test Date:**	19/04/202
Test case Version:	1	**Use Case Reference(s):**	Mosquito Image Detection
Revision History:	v1.0 – Initial AI detection model implemented in FYP-I v2.0 – Detection testing process structured and documented in FYP-II v2.1 – Improvements made to detection accuracy and response handling		
Objective	To verify that the system detects mosquito images correctly		

Product/Ver/Module:	Dengue Monitoring System – AI Detection	
Environment:	Web application with AI model	
Assumptions:	Valid image is provided	
Pre-Requisite:	User is logged in	
Step No.	**Execution description**	**Procedure result**
1	User opens detection module	Upload interface is displayed
2	User uploads image	Image is received by system
3	User submits image	AI model processes image
4	System returns result	Detection result is displayed
Comments: Detection module functioning properly		
☑ Passed ☐ Failed ☐ Not Executed		

4.3.3 Breeding Site Reporting Module Test Case

Breeding Site Reporting Module			
Reporting Use Case			
Test Case ID:	TC-03	**Test Date:**	19/04/2026
Test case Version:	1	**Use Case Reference(s):**	Breeding Site Reporting
Revision History:	v1.0 – Reporting feature introduced in FYP-I v2.0 – Test case design added and structured in FYP-II v2.1 – Enhancements made to report submission and data storage		
Objective	To verify submission of breeding site reports		
Product/Ver/Module:	Dengue Monitoring System – Reporting		
Environment:	Web application		
Assumptions:	User provides valid input		
Pre-Requisite:	User is logged in		
Step No.	**Execution description**	**Procedure result**	

1	User opens reporting form	Upload interface is displayed
2	User enters details	Image is received by system
3	User submits report	System stores report in database
4	System confirms submission	Success message displayed
Comments: Detection module functioning properly		
☑ Passed ☐ Failed ☐ Not Executed		

4.3.4 Analytical Dashboard Test Case

Analytical Dashboard			
Analytics Use Case			
Test Case ID:	TC-04	**Test Date:**	19/04/202
Test case Version:	1	**Use Case Reference(s):**	Data Analytics Dashboard
Revision History:	v1.0 – Dashboard concept introduced in FYP-I v2.0 – Analytics features implemented and tested in FYP-II v2.1 – Improvements made to data visualization and user interaction		
Objective	To verify analytics data visualization		
Product/Ver/Module:	Dengue Monitoring System – Dashboard		
Environment:	Web application		
Assumptions:	Data exists in database		
Pre-Requisite:	User is logged in		
Step No.	**Execution description**	**Procedure result**	
1	User opens dashboard	Dashboard loads successfully	
2	User selects analytics	Data is retrieved	
3	System displays charts	Graphs and statistics shown	
Comments: Dashboard working as expected			

☑ *Passed* ☐ *Failed* ☐ *Not Executed*

4.3.5 AI Chatbot Module Test Case

AI Chatbot Module			
Chatbot Interaction Use Case			
Test Case ID:	TC-05	**Test Date:**	19/04/2026
Test case Version:	1	**Use Case Reference(s):**	AI Chatbot Interaction
Revision History:	v1.0 – Chatbot module introduced in FYP-II v2.1 – Test case developed and refined for chatbot interaction		
Objective	To verify that the AI chatbot responds correctly to user queries		
Product/Ver/Module:	Dengue Monitoring System – AI Chatbot		
Environment:	Web application with chatbot integration		
Assumptions:	Chatbot service is active and connected		
Pre-Requisite	User is logged in and chatbot interface is accessible		

Step No.	**Execution description**	**Procedure result**
1	User opens chatbot interface	Chat window is displayed
2	User enters a query	Input is received by system
3	System processes query	Chatbot generates response
Comments: Chatbot responding correctly to user queries		
☑ *Passed* ☐ *Failed* ☐ *Not Executed*		

4.4 Test Metrics

Test metrics evaluate how effectively the system performs during testing. These metrics are based on the execution of all test cases.

4.4.1 Mosquito Detection Metrics

Metric	Purpose

Number of Test Cases:	5
Number of Test Cases Passed:	5
Number of Test Cases Failed:	0
Test Case Defect Density:	0
Test Case Effectiveness:	100%
Traceability Matrix:	All requirements covered

4.4.2 AI Chatbot Metrics

Metric	Purpose
Average Response Time	2 seconds
AI Detection Time	2–3 seconds
Report Submission Time	1 second
System Availability	Stable
Error Rate	No major errors

4.4.3 Usability and User Experience Metrics

Metric	Purpose
Usability	Easy to use
User Success Rate	High
Interface Clarity	Clear
Feature Accessibility	All features accessible

4.5 Conclusion

This chapter presented the implementation and testing of the proposed dengue monitoring system. The system was developed using a combination of web technologies and deep learning techniques. The mosquito detection module was implemented using a CNN model, while other modules such as reporting and analytics were integrated into the platform.

Chapter 5: Experimental Results and Analysis

5.1 Introduction

This chapter presents the experimental results and analysis of the proposed AI-Powered Dengue Monitoring and Mosquito Detection Platform. The purpose of this chapter is to evaluate the performance of the system and examine how effectively it meets the objectives defined earlier. The results include the performance of the mosquito detection model as well as the outcomes of system testing.

The analysis is presented in a clear and simple manner using tables and descriptive explanations. This chapter also discusses the strengths and limitations of the system based on the observed results.

5.2 AI Model Results

The results presented in this section evaluate the performance of both the mosquito detection model and the AI chatbot module. The mosquito detection model is assessed using standard classification metrics such as accuracy, precision, and recall, which indicate its ability to correctly identify mosquito images. In addition, sample predictions are included to demonstrate the model's behavior on different inputs. The chatbot model is evaluated based on response accuracy and intent matching, which reflect its ability to understand user queries and generate appropriate responses. Overall, these results provide a clear understanding of how effectively the AI components perform within the system.

Table 2: Performance of Mosquito Detection Model

This table presents the overall performance of the mosquito detection model using standard evaluation metrics.

Metrics	Values
Model Type	Convolutional Neural Network (CNN)
Task	Image Classification (Mosquito / Non-Mosquito)
Accuracy	91%
Precision	89%
Recall	90%
Average Response Time	2-5 sec
Dataset Type	Image dataset (mosquito and non-mosquito samples)
Remarks	High accuracy on clear images; performance slightly decreases on noisy or blurred inputs

Table 3: Performance of AI Chatbot Model

This table shows the performance of the chatbot in handling user queries and generating relevant responses.

Metrics	Values

Model Type	NLP-based Chatbot Model
Task	Text-based Query Handling
Response Accuracy	88%
Intent Matching Accuracy	85%
Average Response Time	< 2 sec
Supported Queries	Dengue symptoms, prevention, general awareness
Remarks	Generates relevant responses; accuracy depends on clarity of user queries

Table 4: Predictions of Mosquito Detection Model

This table provides sample outputs of the mosquito detection model to demonstrate its prediction behavior.

Image Sample	**Actual Label**	**Prediction Label**	**Confidence Score**
Image 1	Dengue Mosquito	Non-Dengue Mosquito	70%
Image 2	Dengue Mosquito	Dengue Mosquito	90%
Image 3	Non-Dengue Mosquito	Non-Dengue Mosquito	95%
Image 4	Dengue Mosquito	Dengue Mosquito	93%
Image 5	Other Object	Other Object	97%

5.3 System Testing Results

The results in this section present the outcomes of system testing performed on all major modules of the platform. The testing process was carried out using predefined test cases to verify whether each component behaves as expected. The results include a summary of all test cases as well as a detailed comparison between expected and actual outcomes. These results confirm the correctness, stability, and proper integration of the system components. The testing also ensured that the system performs reliably under normal user interaction scenarios without unexpected behavior.

Table 5: Summary of Test Case Results

This table provides an overview of all test cases executed for different system modules along with their outcomes.

Test Case	**Module**	**Description**	**Results**
TC-1	User Authentication	Verify user login functionality	Passed
TC-2	Mosquito Detection	Verify image detection using AI model	Passed
TC-3	Breeding Site Reporting	Verify report submission functionality	Passed
TC-4	Analytics Dashboard	Verify analytics display and	Passed

		data visualization	
TC-5	AI Chatbot	Verify chatbot response to user queries	Passed

Table 6: Detailed Test Execution Results

This table compares the expected results with the actual outcomes for each test case.

Test Case	Expected Outcome	Description	Results
TC-1	User logs in successfully	User redirected to Home Page	Passed
TC-2	System detects mosquito correctly	Correct prediction displayed	Passed
TC-3	Report is saved in database	Report stored successfully	Passed
TC-4	Dashboard displays analytics	Charts and data displayed	Passed
TC-5	Chatbot responds correctly	Relevant response generated	Passed

5.4 Performance Evaluation

The performance evaluation of the system focuses on measuring its efficiency, responsiveness, and overall usability during operation. The metrics presented in this section include system response time, AI processing time, and usability observations. These results help assess how well the system performs under normal usage conditions and whether it provides a smooth user experience.

Table 7: System Performance Metrics

This table presents the performance of the system based on response time and operational

Metric	Value
Average System Response Time	4 Sec
AI Detection Time	2-5 Sec
Chatbot Response Time	3 Sec
Report Submission Time	5 Sec
System Availability	Stable during testing
Error Rate	No critical errors observed

Table 8: Usability Evaluation

This table presents observations related to the usability of the system, including ease of use, navigation, and user interaction.

Criteria	Observation
Ease of Use	System is simple and user-friendly

Navigation	Users can easily move between modules
Interface Design	Clean and understandable layout
Feature Accessibility	All features are easily accessible
User Satisfaction	Positive feedback during testing

5.5 Conclusion

This chapter presented the experimental results and analysis of the proposed dengue monitoring system. The performance of the mosquito detection model and chatbot was evaluated using appropriate metrics, and the results indicate that both components perform effectively in their respective tasks. The system testing results confirm that all modules are functioning correctly, while the performance evaluation shows that the system operates efficiently and provides a smooth user experience. Overall, the proposed platform successfully meets its objectives by integrating artificial intelligence with user-friendly features to support dengue monitoring and awareness.

Chapter 6: Conclusion and Future Directions

6.1 Introduction

This chapter presents the overall conclusion of the proposed AI-Powered Dengue Monitoring and Mosquito Detection Platform. It summarizes the work carried out in previous chapters, highlights the key findings and results, and evaluates whether the objectives of the project were successfully achieved. In addition, this chapter discusses the limitations encountered during development and provides recommendations for future improvements of the system.

6.2 Conclusion

The main objective of this project was to develop a system that integrates artificial intelligence with a web-based platform to support dengue monitoring, mosquito detection, and community awareness. The system was designed with multiple modules, including mosquito image detection, breeding site reporting, analytics dashboard, and an AI chatbot for user interaction. The results obtained from the implementation and testing phases show that the system performs effectively in its intended tasks. The mosquito detection model achieved good accuracy and was able to correctly classify most input images. The chatbot module also demonstrated the ability to respond to user queries related to dengue awareness. In addition, the system successfully allows users to report breeding sites and view analytical information through the dashboard.

The scope of the project was largely covered, as all planned modules were implemented and tested. The system meets the defined objectives by providing a platform that combines detection, reporting, and awareness features in a single application. Compared to existing approaches that focus on individual aspects, the proposed system offers a more integrated solution.

However, some challenges were encountered during the development process. One of the main challenges was related to training the AI model with a limited dataset, which can affect the accuracy of predictions in certain cases. Another challenge was ensuring smooth integration between the AI model and the web application. Additionally, variations in image quality and user input can influence system performance. Despite these challenges, the system was successfully developed and tested, demonstrating its potential to assist in dengue monitoring and awareness activities.

6.3 Future Direction

Although the proposed system performs well, there are several areas where it can be further improved and enhanced. One possible improvement is the use of a larger and more diverse dataset for training the mosquito detection model. This would help improve the accuracy and robustness of the model, especially in handling complex or low-quality images. The chatbot module can also be enhanced by integrating more

advanced natural language processing techniques to improve its understanding of user queries and provide more detailed responses. Another improvement would be to integrate real-time data sources, such as weather or environmental data, to support more advanced dengue prediction and risk analysis. This would make the system more useful for early warning and decision-making.

The system can also be extended to support mobile applications, allowing users to access features more conveniently. In addition, location-based services can be integrated to provide better tracking of reported breeding sites. Finally, future work can focus on improving system scalability and deploying the platform in real-world environments to evaluate its performance on a larger scale.

6.4 Conclusion

This chapter summarized the overall work of the project and highlighted the key outcomes and findings. The proposed system successfully integrates artificial intelligence with a user-friendly platform to support dengue monitoring and awareness. While some limitations exist, the system provides a strong foundation for further development. The suggested future improvements can enhance the system's capabilities and make it more effective in real-world applications.

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